



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

UNIT TEST -3, NOVEMBER - 2024

Time Allowed : 1.30 Hours

CHEMISTRY

[Max. Marks : 35]

PART - I

I. Choose the correct answer.

5x1=5

1. In case of alkali metal halides, the ionic character increases in the order.

- a) $MF < MCl < MBr < MI$ b) $MI < MBr < MCl < MF$
c) $MI < MBr < MF < MCl$ d) None of the above

2. **Assertion** : An Ideal solution obeys Raoult's laws.

Reason : In an Ideal solution, solvent - solvent as well as solute - solute interactions are similar to solute - solvent interactions.

- a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion
b) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
c) Assertion is true but Reason is false d) Both Assertion and Reason are false

3. The General formula for Cycloalkanes

- a) C_nH_n b) C_nH_{2n} c) C_nH_{2n-2} d) C_nH_{2n+2}

4. Chemical Formula of washing soda is

- a) $NaCl$ b) $NaOH$ c) $Na_2CO_3 \cdot 10H_2O$ d) $NaHCO_3$

5. Which of the following can be used as a halide component for Friedel - craft reaction?

- a) Chlorobenzene b) Bromo benzene c) Chloro ethene d) Isopropyl Chloride

PART - II

II. Answer any Three questions. Q.No. 11 is compulsory.

3x2=6

6. How is Plaster of paris prepared?

7. Discuss any two similarities between Beryllium and Aluminium.

8. Define the term 'Isotonic Solution'.

9. How does Huckel rule help to decide the aromatic character of a compound?

10. Write note on Ozonolysis.

11. 5.845g of NaCl is dissolved in water and the solution was made upto 500 ml using a standard flask. Calculate the molarity of the solution.

PART - III

III. Answer any three questions. Q.No. 17 Is compulsory.

3x3=9

12. Write the Chemical equations for the reactions involved in Solvay Process of preparation of Sodium Carbonate.
13. Describe briefly the biological importance of Calcium and Magnesium.
14. How do you determine the molar mass of solute from elevation of Boiling point? 53
15. What is vapour pressure of Liquid? What is relative lowering of Vapour Pressure?
16. Explain Markovnikoff's rule with suitable example.
17. i) n - hexane $\xrightarrow[873\text{ K}]{\text{Cr}_2\text{O}_3/\text{Al}_2\text{O}_3}$? (1½)
- ii) n - butane $\xrightarrow[298\text{ K}]{\text{AlCl}_3/\text{HCl}}$? (1½)

PART - IV

IV. Answer all the questions.

3x5=15

18. a) i) Describe briefly the biological importance of Sodium and Potassium. (3)
- ii) Mention the uses of Plaster of Paris. (2)
- (OR)
- b) State Raoult's law and Obtain expression for lowering of Vapour pressure when Non - Volatile solute is dissolved in solvent.
19. a) i) What is Molal depression constant? Does it depend on nature of the solute? (3)
- ii) What is Osmosis? (2) (OR)
- b) How will you effect the following Conversions.
- i) Ethylene $\longrightarrow$ Ethylene glycol. (2)
- ii) Phenol $\longrightarrow$ Benzene (1½)
- iii) Benzene $\longrightarrow$ Toluene (1½)
20. a) i) How is quick lime prepared? (3)
- ii) Why alkaline earth metals are harder than alkali metals? (2)
- (OR)
- b) Explain the structure of Benzene.



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